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A Qualitative Inquiry Into Metacognitive Strategies of Postgraduate Students in Employing ChatGPT for English Academic Writing

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ABSTRACT

Recent years have witnessed a growing application of generative artificial intelligence (GenAI) technology in writing instruction. Students should mobilise their metacognitive strategies during this endeavour to maximise the benefits of GenAI while avoiding the potential negative impacts. Within the context of tertiary education in Hong Kong, this qualitative research investigates 13 postgraduate students' metacognitive strategies when utilising ChatGPT for English academic writing. Using data collected from audio recordings, semi-structured interviews and human–GenAI chat logs, we explored students' metacognitive strategies within five dimensions, namely, planning, monitoring, evaluating, information management and debugging. The identified themes and subthemes within each dimension exhibited similarities to those observed in other educational settings. Meanwhile, certain metacognitive strategies (e.g., debugging) were found to be unique to the context of GenAI-supported academic writing. This study contributes to the literature in two ways. Firstly, it provides a detailed and thorough understanding of metacognitive strategies, complementing the extensive quantitative research in this area. Secondly, this research expands the framework of metacognitive strategies to the context of GenAI-supported academic writing learning.

1 | Introduction

As a unique genre of writing task, academic writing in English refers to the endeavour of 'using English to document scientific knowledge, communicate research processes, describe research outcomes, and outline research implications' (Teng and Yue 2023, 237). It is an intricate, challenging task, particularly for English as a second language (L2) students due to factors such as lack of prior training and the need to handle complex writing tasks (Teng and Yue 2023). These challenges, combined with limited L2 proficiency, make academic writing particularly demanding for these students (Lin and Morrison 2021). To address this concern, various instructional approaches have been applied to sustain the learning of English academic writing for L2 learners. Recently, scholars have posited the pivotal role of metacognitive strategies in promoting academic writing proficiency, as academic writing constitutes a cognitive undertaking wherein students actively engage in metacognitive processes, including planning, monitoring and evaluating (Teng, Qin, and Wang 2022; Teng and Yue 2023). The examination of students' metacognitive strategies in English academic writing offers valuable insights for advancing their writing proficiency and fostering effective self-regulated learning practices. This holds particular significance in contemporary society, as the rapid advancements in computer technology exert a profound influence on and reshape the learning environment of writing education (Barrot 2023; Yan 2023).

With the advent of ChatGPT, a generative artificial intelligence (GenAI)-guided chatbot, there has been a growing focus on

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the utilisation of GenAI technology in L2 education (Huang, Wang, and Zhang 2024; Wang and Xue 2024; Wu, Wang, and Wang 2024), particularly to enhance the learning of academic writing for L2 learners (e.g., Nugroho et al. 2024; Zou and Huang 2024). GenAI tools offer instantaneous, customised feedback on writing, thereby facilitating students' acquisition of knowledge from errors and comprehension of their areas of proficiency and deficiency (Rad, Alipour, and Jafarpour 2023). Besides, recent research has indicated that GenAI tools can address lower-level issues in writing, such as vocabulary and grammar, allowing students to allocate a larger portion of their cognitive capacity towards the development of critical and creative thinking during writing (Zou and Huang 2024). Therefore, scholars acknowledged that various GenAI tools may play a facilitating role in academic writing, particularly for L2 English speakers (Hosseini, Rasmussen, and Resnik 2023). Nevertheless, caution should be taken to employ GenAI technology in academic writing, as concerns regarding the violation of academic integrity have been repeatedly raised by scholars (Barrot 2023; Hosseini, Rasmussen, and Resnik 2023; Yan 2023). Additionally, GenAI tools may include inaccurate information and formulaic expressions in their responses (Nugroho et al. 2024; Zou and Huang 2024). Notably, an overreliance on GenAI tools may stiffen students' critical and creative thinking, potentially yielding adverse effects on their writing learning (Huang and Tan 2023; Zou and Huang 2024).

Given these concerns, students should mobilise their metacognitive strategies which enable them to critically evaluate the learning tasks, engage in self-reflection regarding their academic performance and assess the efficacy of their learning approaches, thus ensuring the appropriate employment of GenAI technology (Zhang and Zou 2022). Therefore, within the context of tertiary education in Hong Kong, this study aims to investigate the metacognitive strategies employed by postgraduate students when utilising ChatGPT for English academic writing. The findings of this study extend the current research on metacognitive strategies to the domain of GenAI-supported academic writing learning, providing reference for future research in this area.

2 | Literature Review

2.1 | Metacognition and Metacognitive Strategies in L2 Writing

Metacognition, often described as learners' understanding and utilisation of cognitive processes (Flavell 1979, 1981), has been widely employed as a theoretical framework for research on L2 learning and teaching (e.g., Teng, Qin, and Wang 2022; Teng and Ma 2024; Teng and Yue 2023; Zhang and Zhang 2019). As an umbrella term, metacognition encompasses various essential components. While different definitions have been used in the literature, scholars generally agree that metacognition consists of metacognitive knowledge, metacognitive experience and metacognitive strategies (Efklides 2006, 2008; Lee and Mak 2018; Teng, Qin, and Wang 2022; Teng and Zhang 2024; Zhang and Zhang 2019). Metacognitive knowledge primarily refers to 'learners' knowledge or beliefs about what factors or variables act and interact in what ways to affect the course and outcome of cognitive enterprise' (Flavell 1981, 5). Metacognitive experience pertains to students' consciousness and emotions when they are required to handle information associated with an assignment (Efklides 2006; Zhang and Zhang 2019). Metacognitive strategies refer to the skills employed by students to monitor, regulate and control their own thinking and learning (Lee and Mak 2018; Teng and Qin 2024). The three components are closely interconnected, 'with metacognitive knowledge being an indispensable component and metacognitive strategies playing a key role in metacognitive control, monitoring, and regulation and helping shape metacognitive experiences' (Lee and Mak 2018, 1087).

In the domain of L2 education, research on metacognition predominantly centres on students' employment of metacognitive strategies in the learning of specific language skills, with a particular emphasis on writing learning (e.g., Teng, Qin, and Wang 2022; Teng and Yue 2023; Teng and Zhang 2016). L2 writing has long been acknowledged as a cognitively demanding endeavour for L2 learners (Bui and Kong 2019). The utilisation of diverse metacognitive strategies becomes essential for ensuring the efficacy of L2 writing acquisition, as these strategies empower students to actively engage in the analysis, regulation and direction of their own learning, before, during and after the learning endeavour (Qin and Zhang 2019; Zhang and Zhang 2019). Given the significance of metacognitive strategies in L2 writing, this study focuses on students' metacognitive strategies in the application of GenAI technology for English academic writing.

The subdimensions of metacognitive strategies may exhibit variations contingent upon the employed framework or model. Nonetheless, the following subdimensions are widely acknowledged: planning, monitoring and evaluating (Schraw 1998; Zhang and Zhang 2019; Zhang and Zou 2022). Specifically, planning refers to the process of setting goals, organising tasks and formulating strategies to efficiently tackle a given cognitive task (Broadbent and Poon 2015; Schraw 1998). Monitoring involves the continuous assessment and awareness of one's own cognitive processes, including the monitoring of comprehension, progress and performance of the task (Garcia, Falkner, and Vivian 2018; Teng, Qin, and Wang 2022). Evaluating entails the critical examination and appraisal of one's own performance and outcomes, enabling the identification of strengths, weaknesses and areas for improvement (Schunk 2005; Teng and Yue 2023). Together, these three components constitute a comprehensive framework for metacognitive engagement and self-directed learning. In addition to the three widely recognised metacognitive strategies, a recent empirical study conducted by Teng, Qin, and Wang (2022) with undergraduates at a university in China revealed an additional set of two metacognitive strategies in the domain of English academic writing, namely, information management and debugging strategies. Information management strategies encompass a range of skills that involve the efficient processing, organisation, elaboration and summarisation of information (Nilson 2013). Debugging strategies concern the ability to correct errors in comprehension and performance (Nilson 2013). The aforementioned studies highlight the multifaceted nature of metacognitive strategies in the domain of L2 writing.

Recently, many empirical studies have consistently confirmed the significance of metacognitive strategies on L2 writing learning, such as the heightened level of learning engagement (Bui and Kong 2019) and the enhancement of L2 writing

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performance (Qin and Zhang 2019; Teng, Qin, and Wang 2022; Teng and Yue 2023). However, there are two notable issues in the literature. Firstly, the existing research predominantly adopted quantitative research methods and relied on self-reported questionnaires for data collection. 'Metacognitive strategies are often not as observable ... and so are difficult to identify and assess' (Alvi and Gillies 2023, 683). Thus, scholars have indicated that 'Like many research studies in educational psychology, the use of survey questionnaire has its limitations' (Zhang and Zhang 2013, 112). The adoption of qualitative research methods is justified to gain in-depth insights into the metacognitive strategies employed by L2 writers. With data collected from multiple sources, such approach enhances the reliability and validity of the findings by leveraging the benefits of data triangulation.

Secondly, Zhang and Zhang (2013) argued that metacognition should be regarded as a dynamic system which is 'subject to social, contextual and cultural modifications' (p. 116). Over the past decade, the wide application of computer technology has profoundly transformed the landscape of language acquisition, fostering the process of language learning among students and reshaping their learning strategies (Jiang and Yu 2022; Ranalli 2021). This holds particular merit for the advent of AI-guided chatbots, such as ChatGPT, which may complement teacher feedback and enhance students' academic writing performance (Lu et al. 2024). It is conceivable that students may employ metacognitive strategies that bear resemblances to, yet also exhibit disparities from, the metacognitive strategies elucidated in the literature (Teng, Qin, and Wang 2022; Zhang and Zhang 2019; Zhang and Zou 2022). Therefore, there is a pressing need for empirical investigation to explore students' employment of metacognitive strategies in the new context of GenAI-supported writing learning.

2.2 | Students' Application of GenAI Technology in Writing Learning

Recently, the progress of GenAI has engendered keen interest among educators and scholars to apply this cutting-edge technology in the domain of writing education (Barrot 2023; Barrett and Pack 2023; Guo et al. 2024; Su, Lin, and Lai 2023). Empirical studies have investigated the application of GenAI technology by students as a means to enhance their writing performance. For instance, Zou and Huang (2024) found that postgraduate L2 English writers at Chinese universities integrated ChatGPT into their entire writing process. Specifically, the students effectively employed ChatGPT to generate ideas, create outlines, revise the content and proofread their manuscripts. In a separate study that involved undergraduate students at universities in Indonesia, Nugroho et al. (2024) observed that students employed ChatGPT to facilitate prewriting planning and organisation. Additionally, students also utilised ChatGPT throughout the writing process to aid in vocabulary selection and syntactical modifications, enhancing their writing proficiency.

The integration of GenAI technology has brought about a range of positive impacts on students' writing learning. To name a few, owing to the timely provision of personalised feedback by GenAI, students exhibited enhanced writing proficiency subsequent to the application of GenAI technology (Marzuki, Rusdin, and Indrawati 2023; Rad, Alipour, and Jafarpour 2023). Besides, the GenAI technology may lessen the intensity of negative emotions, such as anxiety, which are frequently encountered in writing learning (Guo et al. 2024). Foremost, the application of GenAI can influence various dimensions of self-regulated writing learning, including emotional, cognitive, behavioural and metacognitive aspects (Zhang, Zou, and Cheng 2023). With 30 L2 English writers at a university in Hong Kong, Zhang, Zou, and Cheng (2023) reported that GenAI technology can contribute to the planning phase of writing by offering recommendations and guidance for the writing task and stimulating self-evaluation by catering to students' progress in learning. As planning and evaluation are two important subdimensions of metacognition, GenAI can positively influence metacognition process 'by supporting learner application of planning and self-evaluation' (Zhang, Zou, and Cheng 2023, 14).

Nevertheless, the application of GenAI technology in writing also engenders several negative impacts. The presence of inaccurate information within the output generated by GenAI diminishes students' trust in technological reliability (Nugroho et al. 2024; Zou and Huang 2024). Academic integrity is another concern frequently expressed by students (Nugroho et al. 2024; Yan 2023; Zou and Huang 2024). They often fear that the wide application of GenAI in writing may lead to the violation of academic rules, such as plagiarism. In particular, excessive dependence on GenAI technology can be detrimental to 'creative and critical thinking and the ability to make independent judgments about the quality of writing' (Huang and Tan 2023, 1151). Corroborating this point, postgraduate L2 English writers at Chinese universities identified learning loss as the most eminent adverse outcome resulting from the application of GenAI technology, primarily due to the reduced level of thinking ability (Zou and Huang 2024).

Given the increasing application of GenAI technology in writing, as well as the positive and negative ramifications, it is imperative to investigate students' metacognitive strategies during this undertaking for the following justifications. Firstly, as a pioneering technology, GenAI must be judiciously employed (Nugroho et al. 2024; Yan 2023; Zou and Huang 2024). Students should critically evaluate the reliability of the information generated by GenAI, while concurrently avoiding an undue reliance on this technology. By doing so, they can optimise the advantages and mitigate the potential adverse consequences associated with the application of GenAI for writing. These endeavours can be greatly facilitated by the implementation of metacognitive strategies, which prompt students to 'examine the learning tasks at hand, reflect on their academic performance, and judge the effectiveness of the learning methods they used', thereby enabling students to self-regulate their learning activities in this new educational context (Zhang and Zou 2022, 14). Secondly, empirical inquiry into students' usage of metacognitive strategies within the context of GenAI-supported writing learning is extremely limited. While one recent study has examined the influence of GenAI technology on students' metacognitive strategies (Zhang, Zou, and Cheng 2023), the authors primarily focused on two subdimensions, namely, planning and self-evaluation. In light of the significance of metacognitive strategies in GenAI-supported writing learning, there is a pressing need to conduct a thorough investigation into students' utilisation of various subdimensions of metacognitive strategies as outlined in the literature (Teng, Qin, and Wang 2022; Zhang

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and Zhang 2019; Zhang and Zou 2022). Therefore, using qualitative research methods, this research primarily intends to address the following question: How do students employ various metacognitive strategies (i.e., planning, monitoring, evaluating, information management and debugging) in the application of GenAI technology in English academic writing?

3 | Research Method

3.1 | Context and Participants

This study was conducted during the spring semester of 2024 at a public research university in Hong Kong. The institution demonstrated a proactive stance in promoting the judicious utilisation of GenAI technology for educational purposes. In the autumn of 2023, the university provided accessible ChatGPT accounts to all enrolled students and faculty members. Following one semester of implementation, students became familiar with employing ChatGPT as a means to enhance their learning experiences.

The research was carried out at the 13-week subject specifically designed to impart research methodologies in linguistic studies taught by the first author of the paper. Invitations were sent to all 44 students who were registered for the subject, and 13 of them volunteered to participate in this research (Table 1). They were enrolled in diverse Master's programmes within the field of humanities, encompassing disciplines such as linguistics, literature and language education. All participants were L2 English learners, with an average score of 6.5 on the International English Language Testing System (IELTS), indicating an above-medium level of English proficiency.

According to the curriculum, students were required to compose a 2000-word introduction and literature review pertaining to a self-selected research topic of interest at the end of the semester. Throughout the duration of the subject, the students were encouraged to leverage the capabilities of ChatGPT to facilitate their writing endeavours. However, all work submitted by students for assignments must be their own original creations. The use of ChatGPT is solely intended for revisions and generating ideas. Additionally, the students must declare and properly reference any use of GenAI tools in their submissions. Notably, the students were allowed the autonomy to employ either the ChatGPT provided by the university or other versions of ChatGPT available online. Participation in the study was voluntary and uncompensated. Prior to the study, ethical approval was obtained from the Institutional Review Board.

TABLE 1 | Demographic information of participating students.

Pseudonym	Gender	Age	IELTS score
Max	Male	23	6.5
Lucy	Female	23	6.5
Zoe	Female	25	7.5
Nora	Female	24	6.5
Lily	Female	23	6.5
Wendy	Female	24	6.0
Jessica	Female	22	6.5
Evelyn	Female	23	6.5
Leah	Female	25	6.0
Faye	Female	26	6.0
Kaitlyn	Female	34	6.5
Tia	Female	22	6.5
Yvonne	Female	24	7.0

3.2 | Data Collection

We collected data from multiple sources to leverage the benefits of data triangulation in providing in-depth analyses of students' metacognitive strategies. As part of the subject requirement, the students were required to include the following components upon submission of each assignment: screenshots capturing the entirety of the human–GenAI chat log and a 2-min audio recording of their experiences and application of ChatGPT. The human–GenAI chat log delineates the inquiries and instructions posed by the students to ChatGPT, thereby furnishing an objective exposition of the students' application of this technology. Besides, metacognitive strategies reside in working memory and constitute declarative knowledge, which can be verbally expressed as it is retained in short-term memory (Lam 2008). Therefore, shortly after the completion of the writing assignments, the students articulated their experiences and application of ChatGPT in the audio recordings. The analyses of these audio recordings provide a lens through which to understand the students' usage of metacognitive strategies within this undertaking.

One week after the completion of the subject, the students were invited to participate in individual, semi-structured interviews (please refer to Appendix A for the interview protocol). Prior to the interviews, the students were provided with their writing assignments and the human–GenAI chat logs to facilitate the recollection of their application of ChatGPT. The interviews were conducted in Chinese, the participants' native language, and were video recorded. Subsequently, two research assistants at the graduate level transcribed the video recordings into written documents. Prior to data analysis, the transcriptions were sent back to the participating students to check whether there were any inaccuracies. Only the relevant portions pertinent to this study were translated into English for further analysis.

3.3 | Data Analysis

Following the qualitative research on metacognitive strategies (Ku and Ho 2010), we first determined the unit in the data which refers to a segment that 'contains a complete idea' (p. 258). We developed the coding scheme based on the literature (Teng, Qin, and Wang 2022; Zhang and Zhang 2019; Zhang and Zou 2022). As shown in Table 2, five dimensions of metacognitive strategies were included in the coding scheme, namely, planning, monitoring, evaluating, information management and debugging. Given the potential distinctive

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TABLE 2 | Coding scheme and samples.

Dimension	Theme	Subtheme	Sample
Planning	Setting goals in the application of ChatGPT for writing	Correcting grammatical errors	I wanted ChatGPT to revise the language of my writing, correcting the grammatical errors. (Tia, interview response)
		Changing writing style	I hoped that ChatGPT could make the writing style of my article more like that of an academic paper. (Max, interview response)
		Providing ideas	I think that with its vast database, ChatGPT can provide me with some unexpected ideas on certain viewpoints or theories. (Lucy, interview response)
	Organising sources available in writing	Understanding the operation mode of ChatGPT	I looked into how ChatGPT operates and have also seen examples shared by others about using it to assist with academic writing. (Lucy, interview response)
		Learning how to provide prompts	'I read some guides on how to effectively provide prompts to get better answers on the internet'. (Faye, interview response)
	Formulating strategies to accomplish the writing task	Using ChatGPT to polish the language	I planned to first write an article in my own language style, then hand it to ChatGPT to check for wording and grammatical errors ... (Max, interview response)
		Revising the language prior to the usage of ChatGPT	To use ChatGPT effectively, I made multiple revisions using DeepL to the text before inputting it into ChatGPT. (Jessica, interview response)
Monitoring	Monitoring the progress of the writing task	Monitoring the performance of ChatGPT	As I worried that it (ChatGPT) might produce errors or misinterpret my original intent, I provided it with my text in segments ... (Evelyn, interview response)
		Avoiding plagiarism while ensuring originality	I think it's very difficult to define GenAI plagiarism, so I try to avoid this issue by making some minor changes ... (Jessica, interview response)
Evaluating	Evaluating the roles of human writer and ChatGPT	—	My article needs to be created by myself ... The machine only serves a proofreading function for me. (Max, audio recording)
	Identifying the strengths of ChatGPT	Efficiency	ChatGPT allowed me to quickly obtain and summarise information. (Kaitlyn, interview response)
		Effective in expanding the content	'ChatGPT indeed expanded some of the content in the text ....' (Leah, interview response)
		Useful in editing structure	I feel that ChatGPT would be quite useful in terms of writing structure. (Nora, interview response)
		Language revision	I think the advantage of GPT is that the language it helps to polish is indeed much better ... (Wendy, interview response)
	Identifying the weaknesses and areas for improvement	Miscomprehension of prompts	ChatGPT sometimes cannot fully understand my instructions. (Faye, audio recording)
		Unreliable information	The comments provided by ChatGPT are not always correct. (Nora, interview response)
		Ineffective in editing content	ChatGPT is not very helpful in terms of content for writing literature reviews. (Yvonne, interview response)
		Changing the structure	ChatGPT arbitrarily modified the structure of my article. (Max, audio recording)
		Esoteric vocabulary	Some of the vocabulary ChatGPT used was overly academic and obscure. (Lily, interview response)
		Impersonal writing style	ChatGPT is just a machine; the things it writes lack emotion and warmth, and the style is often hard for people to accept. (Leah, audio recording)

(Continues)

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TABLE 2 | (Continued)

Dimension	Theme	Subtheme	Sample
Information management	Organising and summarising information for the writing task	Actively using ChatGPT	I used ChatGPT to summarise information ... (Kaitlyn, interview response)
		Resisting the usage of ChatGPT	I didn't let ChatGPT to process the information because I believe this is something humans should do, not machines. (Max, audio recording)
	Elaborating on information for the writing task	—	Some research methods were difficult for me to understand, so I asked ChatGPT to explain what these methods were. (Tia, interview response)
	Examining the trustworthiness of information	—	I just wanted to know if it could really understand my article or if it was just making things up ... (Lily, interview response)
Debugging	Correcting errors in information generated by ChatGPT	Refining prompts	I found that sometimes it doesn't seem to respond according to my prompts ... so I changed the prompts and ask again. (Nora, interview response)
		Rewriting	ChatGPT came up with its own interpretation ... so, when I found it, I corrected it back. (Yvonne, interview response)
		Referring to additional tools	It (ChatGPT) tended to give me content which was not what I wanted. So, I used other tools like Google. (Jessica, interview response)

features of metacognitive strategies within the new context of GenAI-supported writing learning, the coding scheme remained adaptable throughout the analysis. Employing a deductive approach initially, we subsequently integrated an inductive approach to identify the previously unexplored patterns and themes that emerged during data analysis (Alvi and Gillies 2023). For example, we identified a new theme, 'evaluating the roles of human writer and ChatGPT', within the dimension of evaluating (Table 2).

Using NVivo 10, the data were coded by two graduate research assistants independently. The data coding followed an iterative process wherein the research assistants repeatedly went forward and backward in reading, coding and developing themes that emerged from the data analysis. The two coders initially coded the audio recordings and interview responses of two students. Any discrepancies were resolved through a process of discussion and consensus. Subsequently, they finished the coding of the remaining data. The intercoder reliability reached 84.21% ($p < 0.05$). As shown in Table 2, the themes within each dimension of metacognitive strategies were established based on the literature. Several subthemes were also identified within the framework of metacognitive strategies. For example, within the dimension of planning, under the theme of 'setting goals in the application of ChatGPT for writing', we identified the following subthemes, 'correcting grammatical errors', 'changing writing style' and 'providing ideas'.

4 | Results

Data analyses revealed the diverse metacognitive strategies utilised by the students when employing ChatGPT in English academic writing. The results related to each of the five dimensions of metacognitive strategies are presented individually.

4.1 | Planning

Three themes were identified within the dimension of planning. Concerning the first theme, setting goals in the application of ChatGPT for writing, three subthemes emerged, including 'correcting grammatical errors', 'changing writing style' and 'providing ideas'. Most students indicated that their primary goal of using ChatGPT was to assist in correcting grammatical errors in their writing. For example, Zoe responded to the interview that 'My grammar is quite poor. I would use two verbs in a sentence or have a sentence without a predicate verb. I want my grammatical errors to be reviewed and corrected by ChatGPT'. Changing their writing style to resemble academic writing was also a goal for the students. For instance, Wendy noted that 'My original manuscript did not sufficiently resemble academic writing. It was not authentic, so I wanted to have it polished' (audio recording). Furthermore, some students articulated that their objective in using ChatGPT was to generate innovative ideas during the writing process. For example, 'I think that with its vast database, ChatGPT can provide me with some unexpected ideas on certain viewpoints or theories' (Lucy, interview response).

In terms of the second theme, organising sources available in writing, we identified two subthemes, namely, 'understanding the operation mode of ChatGPT' and 'learning how to provide prompts'. The students utilised Google and various social media platforms to gather information regarding the operational mechanisms of ChatGPT and to learn how to effectively formulate prompts to elicit optimal responses. For example, Lucy responded to the interview that 'I looked into how ChatGPT operates and have also seen examples shared by others about using it to assist with academic writing'. Faye said, 'I read some guides on how to effectively provide prompts to get better answers on the internet' (interview response).

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We observed noteworthy findings regarding the third theme, formulating strategies to accomplish the writing task. On one hand, most students actively employed ChatGPT to revise the language of their writing by first completing the entire manuscript and then submitting it to ChatGPT for review (subtheme, 'using ChatGPT to polish the language'). Max indicated that 'I planned to first write an article in my own language style, then hand it to ChatGPT to check for wording and grammatical errors, and then proceeded with modifications to the writing style' (interview response). On the other hand, some students employed additional tools, such as *DeepL*, to revise their manuscripts before submitting them to ChatGPT (subtheme, 'revising the language prior to the usage of ChatGPT'). Jessica said,

> To use ChatGPT effectively, I made multiple revisions using *DeepL* to the text before inputting it into ChatGPT. I corrected any grammatical errors, expression deficiencies, unclear points of view, and structural issues that I could identify, to prevent it (ChatGPT) from being distracted from these detailed points.
>
> (interview response)

4.2 | Monitoring

Within the dimension of monitoring, we identified a single theme: monitoring the progress of the writing task which encompassed two subthemes: 'monitoring the performance of ChatGPT' and 'avoiding plagiarism while ensuring originality'. During the writing process, owing to the distrust towards the GenAI technology, students constantly monitored the performance of ChatGPT. Evelyn reported in the interview that 'As I worried that it (ChatGPT) might produce errors or misinterpret my original intent, I provided it with my text in segments. I only proceeded if I felt satisfied with the response'. The screenshot of the chat log revealed that Evelyn submitted her writing to ChatGPT paragraph by paragraph (Figure 1).

Another notable issue was that students were aware of the risk of plagiarism caused by utilising ChatGPT. For instance, Jessica indicated in the interview that 'I think it's very difficult to define GenAI plagiarism, so I try to avoid this issue by making some minor changes to the contexts generated by ChatGPT. This way, it won't lead to the possibility of plagiarism'. Meanwhile, some students highlighted the significance of maintaining originality in their writing. Max noted,

> When using it, I want my article to maintain its originality. Words or sentences in the article can be replaced by ChatGPT, but the meaning I intend to convey should remain the same as my original intention. I don't want GenAI to change my meaning.
>
> (interview response)

4.3 | Evaluating

The dimension of evaluating comprised the most fruitful themes and subthemes, probably because the students had utilised ChatGPT or other GenAI tools for nearly one academic year. Thus, they were able to evaluate the strengths and weaknesses of applying this novel technology in their writing. The first identified theme was the evaluation of the roles of the human writer and ChatGPT. All students concurred that they should assume a decisive role throughout the entire writing process, whereas ChatGPT should function solely as a writing assistant, primarily responsible for addressing linguistic issues in their work. For instance, Max noted in the audio recording that 'My article needs to be created by myself ... rather than having these things done by a machine. The machine only serves a proofreading function for me'. Leah also noted that 'Given the current level of development of ChatGPT, it is far from being able to replace human writing ... GenAI tools can only serve as supplementary references and tools for learning' (interview response).

Besides, as shown in Table 2, the students identified various strengths (e.g., efficiency and language revision) and weaknesses (e.g., miscomprehension of prompts, unreliable information, esoteric vocabulary and impersonal writing style) of ChatGPT in the application of academic writing. Interestingly, some students perceived ChatGPT as advantageous for editing content and structure. For instance, 'ChatGPT indeed expanded some of the content in the text, and some of the perspectives and ideas were things I hadn't originally thought of' (Leah, interview response)

FIGURE 1 | Evelyn's submission of writing to ChatGPT.

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and 'I feel that ChatGPT would be quite useful in terms of writing structure' (Nora, interview response). Nevertheless, other students held contrary views, such as 'ChatGPT is not very helpful in terms of content for writing literature reviews' (Yvonne, interview response) and 'ChatGPT arbitrarily modified the structure of my article' (Max, audio recording). The complexity in the perceptions of ChatGPT revealed that, rather than blindly trusting the tool, the students were capable of evaluating its strengths and weaknesses based on their own applications.

4.4 | Information Management

Three themes were identified within the dimension of information management. Regarding the first theme, which involves organising and summarising information for the writing task, the students also exhibited variations in their use of ChatGPT. Some students refused to utilise ChatGPT to process the information, primarily due to their belief that human beings should take responsibility in this endeavour. For example, 'I didn't let ChatGPT to process the information because I believe this is something humans should do, not machines' (Max, audio recording). However, owing to the challenging task of reading academic literature, some students relied on ChatGPT to help them process the information. For instance, as shown in Figure 2, Kaitlyn requested ChatGPT to summarise the ideas of the literature. She reported in the interview that

> I used ChatGPT to summarize information. For assignments like this one, reading lengthy English literature was challenging for me, particularly when it involved some educational theories. So, I directly inputted the content into ChatGPT and let it help me summarize.

The second theme concerned the elaboration of information for the writing task. Most students recognised ChatGPT's capacity to address a wide range of questions. Consequently, they frequently requested it to provide detailed explanations on specific aspects of their writing. For instance, Tia responded in the interview that 'Some research methods were difficult for me to understand, so I asked ChatGPT to explain what these methods were'. Besides, as shown in Figure 3, Leah required ChatGPT to give a definition of itself in the context of her research.

Meanwhile, as reported in the previous sections, the students recognised that some information generated by ChatGPT was unreliable. Therefore, they constantly examined the trustworthiness of the information generated by ChatGPT, which emerged as a unique theme within the information management dimension in the context of GenAI-supported writing learning. For example, after submitting a paragraph to ChatGPT, Lily inquired whether it could comprehend the meaning of the submitted text (Figure 4). She noted that 'I just wanted to know

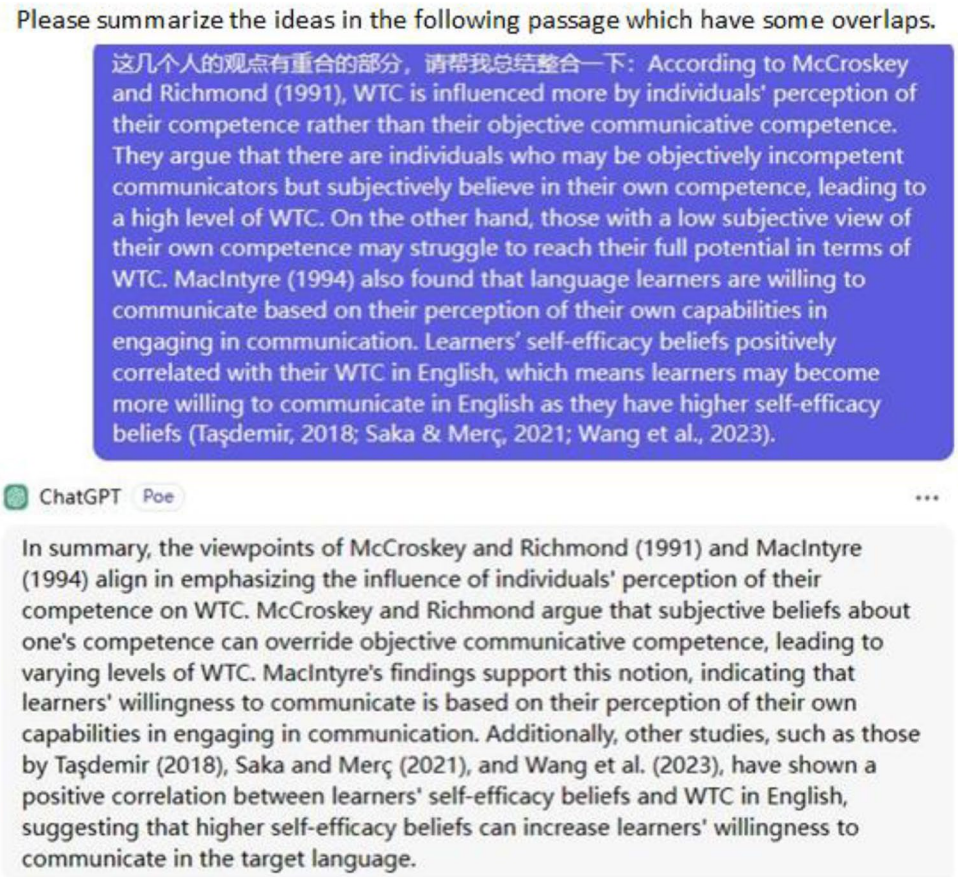

FIGURE 2 | Kaitlyn's request for ChatGPT to summarise the ideas of the literature.

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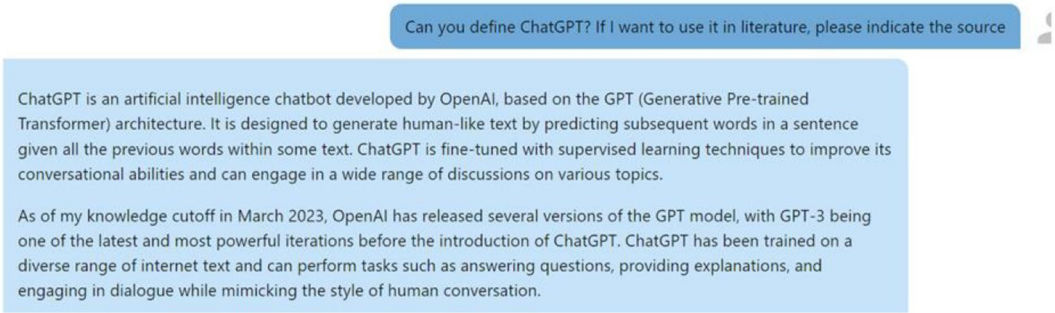

FIGURE 3 | Leah's request for ChatGPT to provide definition.

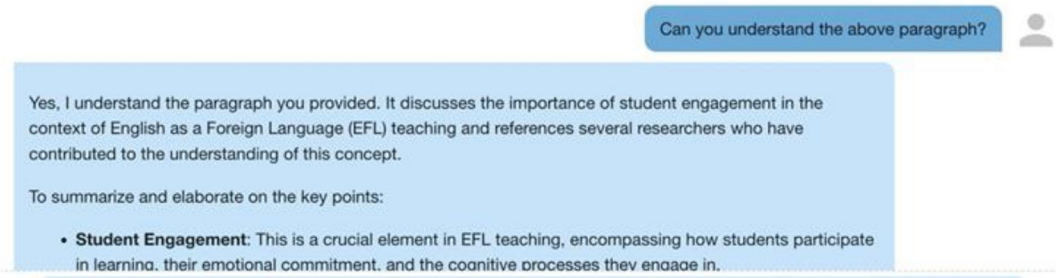

FIGURE 4 | Lily's examination of the trustworthiness of ChatGPT.

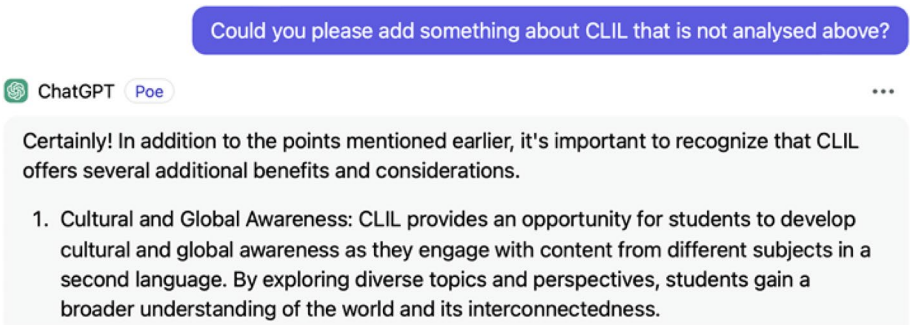

FIGURE 5 | Wendy's additional request for ChatGPT.

if it could really understand my article or if it was just making things up. So, I directly asked it, 'Did you understand the paragraph I wrote?' And GPT's answer was affirmative' (interview response).

4.5 | Debugging

Within the dimension of debugging, the students' metacognitive strategies centre on the correction of errors in the information generated by ChatGPT. Three subthemes emerged: 'refining prompts', 'rewriting' and 'referring to additional tools'. When students detected errors in the information generated by ChatGPT, the most often mentioned strategy was to refine the prompts and make additional inquiries. For instance, Nora noted that 'I found that sometimes it doesn't seem to respond according to my prompts ... so I changed the prompts and ask again' (interview response). The screenshot of the chat log also revealed that Wendy made additional inquiries to ChatGPT after receiving responses she was not satisfied with (Figure 5).

In addition, the students would not blindly accept the information provided by ChatGPT; rather, they would rewrite the contents. Yvonne reported that 'ChatGPT came up with its own interpretation ... so, when I found it, I corrected it back' (interview response). Finally, the students would utilise additional tools, such as Google or other computer apps, for the correction of errors in the information generated by ChatGPT. Jessica noted that 'It (ChatGPT) tended to give me content which was not what I wanted. So, I used other tools like Google' (interview response).

5 | Discussion

The data analysis results revealed various metacognitive strategies employed by postgraduate students when applying ChatGPT

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for L2 English academic writing. As expected, the identified metacognitive strategies in the context of GenAI-supported writing exhibited similarities to those observed in other educational settings. Meanwhile, certain metacognitive strategies were found to be unique to the new context. As demonstrated in the subsequent discussions, a notable characteristic is the intricate interweaving of various dimensions of metacognitive strategies. This interconnection underscores the close relationship between different aspects of metacognition and metacognitive strategies (Lee and Mak 2018; Teng and Yue 2023).

5.1 | Planning: Goals, Resources and Execution

Planning involves setting goals, organising tasks and formulating strategies to efficiently address a specific cognitive task (Broadbent and Poon 2015; Schraw 1998). Within the dimension of planning, the students articulated clear goals for employing ChatGPT, which included correcting grammatical errors, providing ideas and changing writing style. As observed in the existing studies, GenAI technology has demonstrated its remarkable efficacy in revising the language of student writing (Nugroho et al. 2024; Zou and Huang 2024). Besides, due to its capacity for natural language processing, GenAI can understand the requirements of the writing task and provide some innovative points for student writing (Barrett and Pack 2023; Zou and Huang 2024). Therefore, the students in this study reported that they aimed to use ChatGPT to correct grammatical errors and provide ideas for their writing. Notably, the students intended to change the style of their writing to align with the conventions of academic writing. This inclination likely stems from the recognition that academic writing constitutes a distinct genre, characterised by specific features at various levels, including vocabulary and sentence structure (Lin and Morrison 2021). For students who are not well-versed in this genre, it is understandable that, after observing ChatGPT's proficiency in language revision, they would turn to the tool to enhance the academic quality of their writing.

Still, within the dimension of planning, the students actively organised various online resources to understand the operational mechanisms of ChatGPT and to learn how to effectively formulate prompts prior to utilising the tool. Although Yan (2023) observed that students could readily acquire the fundamental skills necessary to utilise ChatGPT for writing, it is reasonable to assume that, in the contemporary information age where diverse information is widely accessible, students are inclined to curate available resources to optimise their use of innovative learning tools. Regarding the third theme, which involves formulating strategies to accomplish the writing task, the students typically completed the entire manuscript before submitting it to ChatGPT for revision. This approach aligns with the argument that GenAI tools should serve as facilitators for academic writing rather than being overly relied upon by students (Huang and Tan 2023; Zou and Huang 2024). Interestingly, despite the efficacy of ChatGPT in revising and editing student writing, some students utilised additional tools, such as *DeepL*, to refine their work before submitting it to ChatGPT. This endeavour reflects students' cautiousness in applying this novel tool to their writing process (further discussed in the following section).

5.2 | Monitoring: Performance Evaluation and Originality Safeguards

Monitoring entails the ongoing awareness of one's cognitive processes, such as tracking progress and task performance (Garcia, Falkner, and Vivian 2018; Teng, Qin, and Wang 2022). The inclusion of incorrect information in the output produced by GenAI may reduce students' confidence in the technology's reliability (Nugroho et al. 2024; Zou and Huang 2024). Therefore, the students in the present study consistently monitored the performance of ChatGPT. Their primary strategy involved submitting their writing to ChatGPT in segments, typically on a paragraph-by-paragraph basis. They would only advance to the next segment once they were satisfied with ChatGPT's responses to their writing.

Furthermore, academic integrity is a prevalent concern highlighted in the literature (Nugroho et al. 2024; Yan 2023; Zou and Huang 2024). The students in the current study also expressed apprehension that the widespread use of ChatGPT in writing could result in violations of academic integrity, especially in terms of plagiarism. Consequently, they underscored the importance of avoiding plagiarism and ensuring originality of their work. For this purpose, the students would make minor revisions to the texts generated by ChatGPT, thereby maintaining the authenticity of their writing.

5.3 | Evaluating: Human–Computer Roles, and Strengths and Weaknesses of ChatGPT

Evaluating involves critically assessing and reviewing the performance and results, allowing for the identification of strengths, weaknesses and areas needing improvement (Schunk 2005; Teng and Yue 2023). A distinctive theme was the evaluation of human–computer roles in academic writing. The students demonstrated a clear understanding of their own roles and the role of ChatGPT in the writing process. They recognised that they should assume responsibility for managing and overseeing the entire writing process, completing the initial draft themselves. ChatGPT, on the other hand, was perceived as an assistant primarily responsible for addressing linguistic issues during the revision stage. This delineation of roles ensures that the students remain actively engaged in the critical thinking and creative aspects of writing while leveraging ChatGPT's strengths in linguistic accuracy and revision (Marzuki, Rusdin, and Indrawati 2023; Zou and Huang 2024). However, it is noteworthy that certain students still employed ChatGPT during the writing process to organise and summarise information for the writing task, which is further discussed in the following section.

The students recognised several strengths of ChatGPT in the context of academic writing, including its efficient responsiveness to human instructions and proficiency in language revision. Meanwhile, they were also aware of the weaknesses of ChatGPT, such as misinterpretation of prompts and the provision of unreliable information, which have been frequently documented by previous studies (Nugroho et al. 2024; Zou and Huang 2024). Notably, the students also identified several weaknesses of ChatGPT that are less frequently reported in the literature, including the use of esoteric vocabulary and an impersonal writing style, which may be attributed to the distinctive

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context of academic writing. As discussed in the previous section, academic writing possesses unique characteristics in terms of vocabulary and sentence structure (Lin and Morrison 2021). Consequently, ChatGPT may suggest the use of certain words or sentences that the students find unfamiliar. Nonetheless, whether these words and the writing style are genuinely problematic or merely perceived as such by the students, despite potentially being appropriate, warrants further investigation in future research. Finally, while some students found ChatGPT effective in editing the content and structure of their writing, others held contrary opinions. Zhang, Zou, and Cheng (2023) observed that the application of GenAI tools can enhance students' evaluative skills during the writing process. It is plausible that the varying experiences of the students in using ChatGPT contributed to their diverse evaluations of its strengths and weaknesses in this study.

5.4 | Information Management: Processing and Examining the Trustworthiness of the Information

Information management strategies include a variety of skills aimed at efficiently processing, organising, elaborating and summarising information (Nilson 2013). Within the dimension of information management, the first emerged theme was to utilise ChatGPT to assist in organising and summarising information when reading academic literature. Organising and summarising key points from academic texts is a challenging task for postgraduate students. Consequently, it is understandable that some students in this study employed ChatGPT to aid them in this endeavour, given its widely acknowledged capabilities in natural language processing. However, reliance on ChatGPT for information processing may have detrimental effects on students' critical and creative thinking skills (Huang and Tan 2023; Zou and Huang 2024). Fortunately, other students resisted using ChatGPT for organising and summarising information, primarily due to their belief that this responsibility should rest with human beings. These students should be praised, and their approach should be recommended to others as well.

Moreover, the students reported that they actively employed ChatGPT to elaborate on information needed for their writing, such as providing explanations for specific concepts in academic writing. While this practice carries the risk of receiving incorrect information generated by ChatGPT (Nugroho et al. 2024; Zou and Huang 2024), the students indicated that they actively examine the trustworthiness of the information through various means, which constitutes a distinct subtheme in their metacognitive strategies. According to the students' descriptions, they exercised discretion in selecting the information generated by ChatGPT. The following section discusses the strategies utilised by students when they actually encountered incorrect information generated by ChatGPT.

5.5 | Debugging: Correcting Errors in Information by Different Means

Debugging strategies represent students' metacognitive control over their learning behaviours, involving adjustments in how material is utilised and how effectively it meets their goals (Nilson 2013). Within this dimension, one major theme was identified: correcting errors in information. While the previous study on students' metacognitive strategies in English academic writing has also identified the dimension of debugging, it primarily focused on correcting errors in comprehension and performance of the writing task itself (Teng, Qin, and Wang 2022). In the context of GenAI-supported academic writing, however, the debugging strategies predominantly pertain to the correction of errors in the information generated by ChatGPT. It was encouraging to observe that when encountering errors in the information generated by ChatGPT, the students actively employed various strategies to correct the errors, including refining prompts, rewriting the contents generated by ChatGPT and applying additional tools. The implementation of these strategies is anticipated to significantly mitigate the potential risks associated with incorrect information. Furthermore, the proactive application of debugging strategies may also enhance students' critical thinking skills when utilising ChatGPT in the writing process (Huang and Tan 2023; Zou and Huang 2024).

6 | Conclusion

With data collected from audio recordings, semistructure interviews and human–GenAI chat logs, this study explores the metacognitive strategies of postgraduate students in the application of ChatGPT for English academic writing. The findings reveal the complexities inherent in various dimensions of metacognitive strategies. This study makes two significant contributions to the existing literature. Firstly, through a qualitative research approach, this study offers an in-depth and comprehensive understanding of metacognitive strategies, complementing the extensive body of quantitative research in this field and providing a reference for the development of scales for future quantitative studies. Secondly, this research extends the framework of metacognitive strategies to the context of GenAI-supported academic writing learning. Given the increasing integration of GenAI technology in educational settings, it is anticipated that various GenAI tools will assume more prominent roles in writing education in the future. Consequently, students should mobilise their metacognitive strategies to effectively navigate their practices in the application of these powerful, yet fallible, tools.

Based on the findings, we propose the following pedagogical suggestions. First, teachers should incorporate training sessions on how to effectively use GenAI tools like ChatGPT into the curriculum. This training should cover setting clear writing goals, formulating effective prompts and understanding the operational mechanisms of GenAI. By doing so, students can maximise the benefits of GenAI tools while maintaining control over their writing process. Second, students should be guided to complete their initial drafts independently before using GenAI for revisions. This approach ensures that students engage deeply with the writing task and use GenAI as a supplementary tool rather than a crutch. Besides, students should be taught to monitor the performance of GenAI tools critically. This includes submitting their work in segments and evaluating the GenAI's feedback before proceeding. Teachers should also emphasise the importance of maintaining academic integrity by making revisions to GenAI-generated content to ensure originality. Finally,

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teachers should instruct students on how to identify and correct errors in GenAI-generated content, including refining prompts, rewriting content and using additional tools to verify information. By fostering these debugging skills, students can mitigate the risks associated with incorrect information and enhance their critical thinking abilities.

Despite the theoretical contribution and the practical suggestions, this study has several limitations that warrant further investigation in future research. Firstly, the sample size was relatively small, consisting exclusively of postgraduate students from a single university, all of whom were majoring in Humanities. Future research should consider recruiting participants from various disciplines and multiple universities to determine whether the findings can be replicated or if new insights may emerge. Secondly, additional variables should be included to further explore students' use of GenAI tools in academic writing. For example, Teng and Yue (2023) found that critical thinking skills significantly impact the relationship between metacognition and writing performance. Given that the use of GenAI tools may affect students' critical thinking skills (Huang and Tan 2023), examining the mediating role of critical thinking skills in the context of GenAI-supported writing could be a promising direction for future research. Lastly, this study primarily employed a cross-sectional design, assessing students' use of metacognitive strategies at a single point in time. Since the application of GenAI tools can influence students' metacognitive strategies (Zhang, Zou, and Cheng 2023), future research should investigate how these strategies evolve and are affected by the use of GenAI tools in writing.

Ethics Statement

The study was approved by the Department of Chinese and Bilingual Studies, The Hong Kong Polytechnic University (HSEARS20240416005).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Appendix A

Interview protocol:

1. What are your goals for utilising ChatGPT in your writing assignment?
2. What are your preparations before the utilisation of this technology in your writing assignment?
3. Have you made any plans prior to the writing process? If yes, what are the plans?
4. Have you been mindful of effectively and appropriately utilising ChatGPT throughout the writing process? If yes, how?
5. When using ChatGPT, how did you use it to help you process, organise, elaborate and summarise the contents of your writing?
6. How did you use it to help you revise the language of the manuscript?
7. Have you identified any errors in ChatGPT's responses? If yes, what did you do?
8. What are the strengths of utilising ChatGPT for your writing assignment?
9. What are the weaknesses of utilising ChatGPT for your writing assignment?
10. If you can use ChatGPT for your writing assignment in the future, what are you going to do to maximise the benefit of this technology?

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